IONET WebApp

IoT Device Management Platform | Backend: api.ionet.sunshineiot.in | v1.0.6

Flutter App

BACKEND STATUS CHECK

Server is LIVE

api.ionet.sunshineiot.in is responding correctly. Returns `{"success": false, "message": "Unauthorized"}` on browser access — this is expected.

Why "Unauthorized"?

The browser sends no JWT token. Every API endpoint requires `Authorization: Bearer <token>`. This is the correct, secure behavior.

App works fine

The Flutter app automatically sends the token with every request after login. No issue in the code. Nothing needs to be fixed.

APP WORKFLOW — STEP BY STEP

1

User Opens App → Auth Screen

User types their **email address** and taps "Send OTP".

`POST /api/v1/auth/request-otp`

Server sends a 6-digit OTP code to the email.

2

OTP Verification

User enters the 6-digit OTP code. App verifies it with the server.

`POST /api/v1/auth/verify-otp`

- 🆕 **New User** → goes to Complete Profile page (name + phone)
- 🔁 **Existing User** → goes directly to Home screen

JWT token is saved locally (SharedPreferences) for auto-login next time.

3

Home Screen Loads

App fetches the user's devices from the server.

`GET /api/v1/devices`

Each device shows: Name, Online/Offline status, Last active time, and Avatar image.
Online = device was seen within **5 minutes**. Otherwise = Offline.

4

User Taps a Device → Chat Opens

Status shows **"Connecting..."**

App opens a **WebSocket** connection to the device:

`wss://api.ionet.sunshineiot.in/ws?deviceId=X&type=mobile&token=JWT`

A **10-second watchdog timer** starts. If device responds → Online ✅. If no response in 10s → Offline ❌.

5

Live Communication via WebSocket

User presses **STATUS** button → app sends: `{"command": "/status"}`

Device replies with JSON sensor data. App parses it and shows in chat as readable text:

```
Relay 1: On
Temperature: 28.5
Fan: Off
```

Device also sends its **available commands list** → shown as quick-tap buttons.

6

User Sends Commands

- ⚡ Tap **Quick Command** chip → instantly sends that command
- ⌨️ Type any **/command** in the text box and press send
- ⚙️ Tap **CONFIG** → opens form to configure device parameters
- 📶 Tap **Wi-Fi Setup** → send SSID and password to device
- 🔁 Tap a **command in chat bubble** → resends it

7

Auto-Reconnect if Connection Drops

If WebSocket disconnects (network issue, device reboot, etc.):

- ⏱️ App waits **3 seconds**
- 🔄 Automatically tries to reconnect
- ⏰ If no message for **10 seconds** → marks device Offline

WEBSOCKET MESSAGE FORMAT

```
APP → DEVICE (you send):
{"command": "/status"} ← request device data
{"command": "/led_on"} ← turn on LED
{"command": "/config", "data": {...}} ← configure device
{"command": "/set-up_wifi", ...} ← send WiFi credentials

DEVICE → APP (device replies):
{"type": "connected"} ← handshake (silent)
{"type": "heartbeat"} ← keep-alive (silent, invisible)
{"type": "device_offline"} ← device disconnected
{"data": {"relay1":"On","temp":28.5}} ← sensor/telemetry data
{"availableCommands": [...]} ← list of valid commands
```

KEY API ENDPOINTS

Action	Method	Endpoint
Send OTP to email	POST	`/api/v1/auth/request-otp`
Verify OTP, get token	POST	`/api/v1/auth/verify-otp`
Get my profile	GET	`/api/v1/users/me`

Create profile (new user)	POST	`/api/v1/users/profile`
Get all devices	GET	`/api/v1/devices`
Add new device	POST	`/api/v1/devices`
Update device name	PUT	`/api/v1/devices/{id}`
Delete device	DELETE	`/api/v1/devices/{id}`
Share device with user	POST	`/api/v1/devices/{id}/share`
Get knowledge channels	GET	`/api/v1/channels`
Get updates/news	GET	`/api/v1/updates`

KEY FEATURES AT A GLANCE

🔑
OTP Login
Email → 6-digit OTP → JWT token auto-saved for next login

📡
Real-time WebSocket
Persistent connection to device. Commands and data flow both ways instantly

⏱️
Watchdog Timer
If device is silent for 10 seconds → automatically marked Offline

🔄
Auto-Reconnect
WebSocket drops? App reconnects after 3 seconds automatically

⚡
Quick Commands
Device sends its command list → shown as tap buttons in chat

👥
Device Sharing
Owner can share device access with other users by email/phone

🌙
Dark / Light Theme
Supports Light, Dark, and System theme. Saved between sessions

📚
Knowledge Channels
PDF guides, YouTube videos, and links — organized in channels